Life Sciences and
Facility Management

Institute of
Computational Life Sciences

Simulation for Beginners

Model Building / Validation and Verification

Day 3

Tatiana Starostina, Marco Brunschwiler,

ICLS: Institute of Computational Life Sciences, 26.03.2026

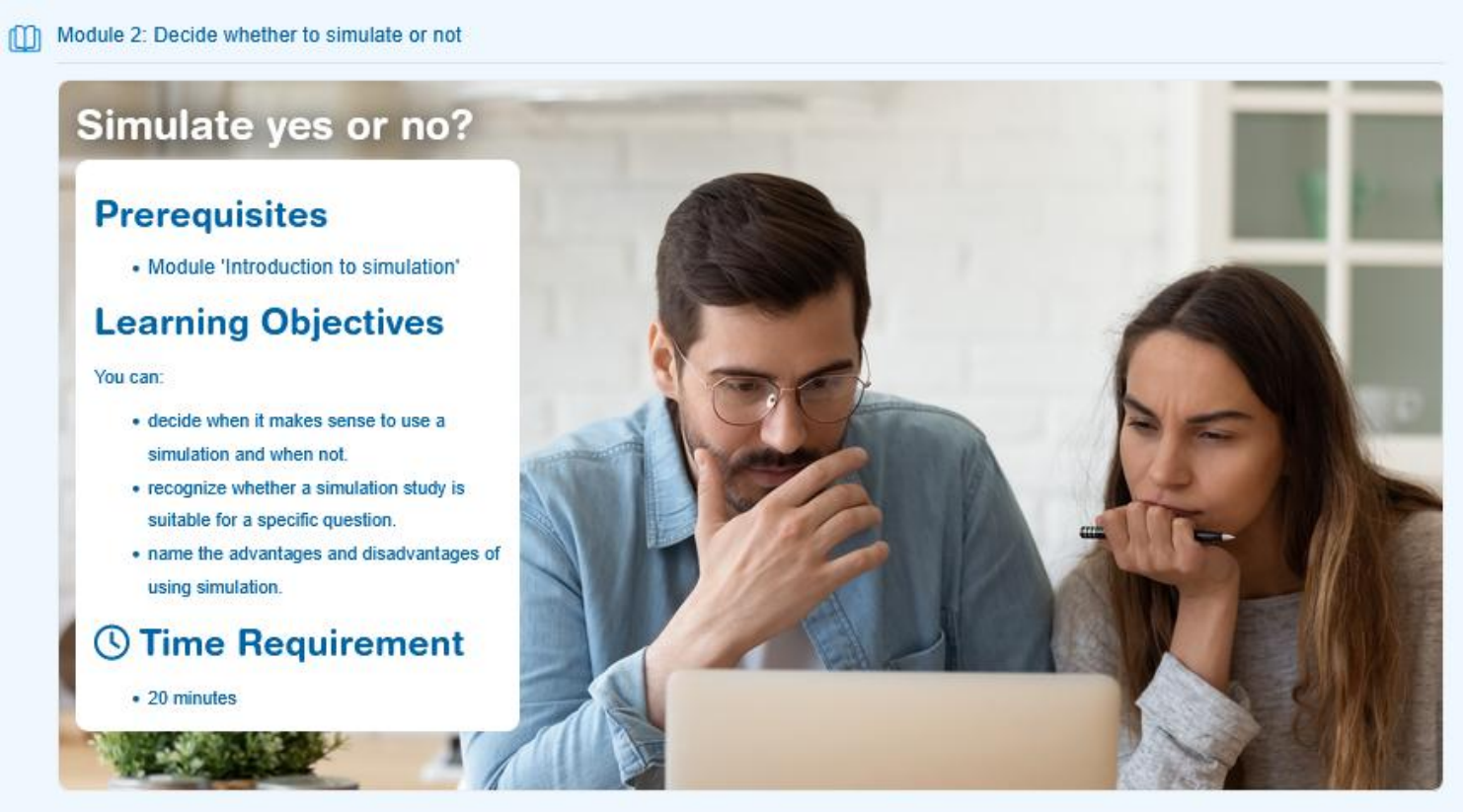
Plan of module

Simulation for Beginners			Marco Brunschwiler / Tatiana Starostina	
CW	**Date**	**Topics**	**Work at home**	**Time expenditure at home**
11	Thursday, 12 March 2026	**Course overview** **Introduction to simulation (Input)** **Introduction to project assignments** **Project assignment 1:** Development of a process flow	**Moodle - Module 1:** What is simulation **Moodle - Module 2:** Installations and first steps **Project assignment 1**	4
12	Thursday, 19 March 2026	**Project example – Vaccination center (20 min)** **Project assignment 2:** conducting experiments and evaluation **Model building:** Build a model from the beginning	**Project Assignments:** Work on Assignments **Moodle - Module 3:** Decide whether to simulate or not	4
13	Thursday, 26 March 2026	**Verification & validation (Input)** **Experiment planning based on research questions (Input)** **Discuss Assignment 1&2 / clarify questions** Model building: Moving resources, state statistics, tally statistics	**Project Assignments:** Work on Assignments **Moodle - Module 4:** Introduction to the simulation study	4
14	Thursday, 2 April 2026	**Easter holidays**	Free	
15	Thursday, 9 April 2026	**Conduct experiments and analyze results** **Discussing results** **Work on Project-Assignments**	**Project Assignments:** Work on Assignments	4
16	Thursday, 16 April 2026	Presentation of results Final review / feedback session	**Project Assignments:** Complete project work	4

Goal: 20 hours

Moodle Module 2

- Additional information and learning control in Moodle course
- **Module 2: Feedback**

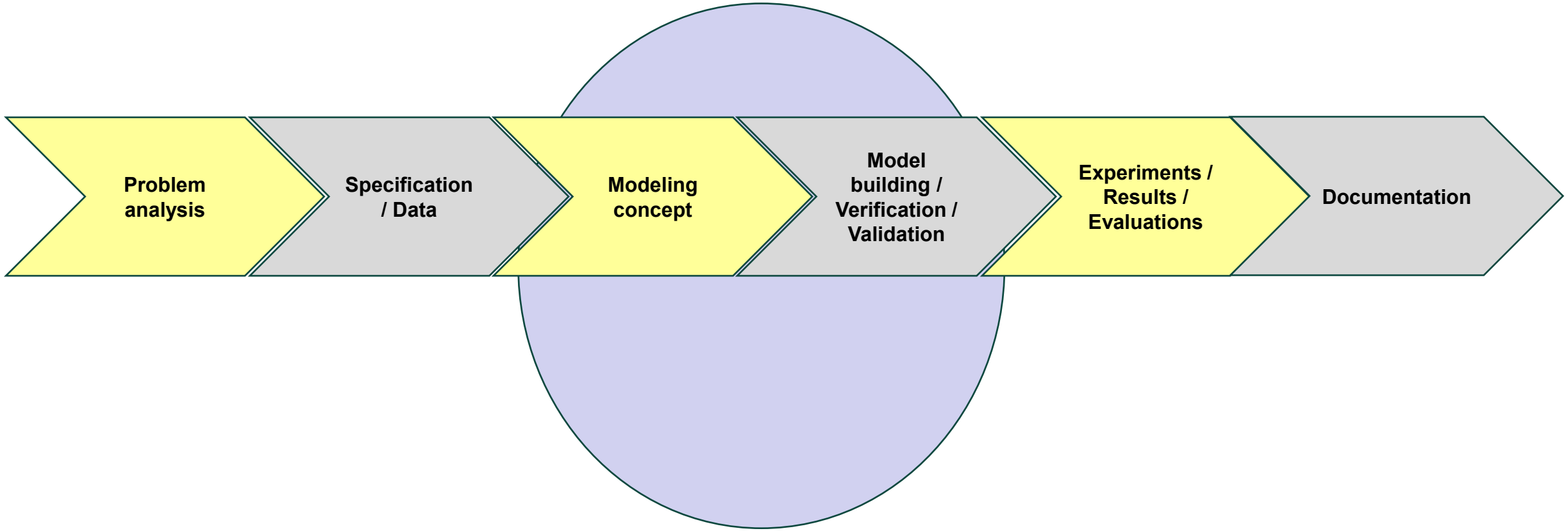


Questions?

Project Assignments
Questions?

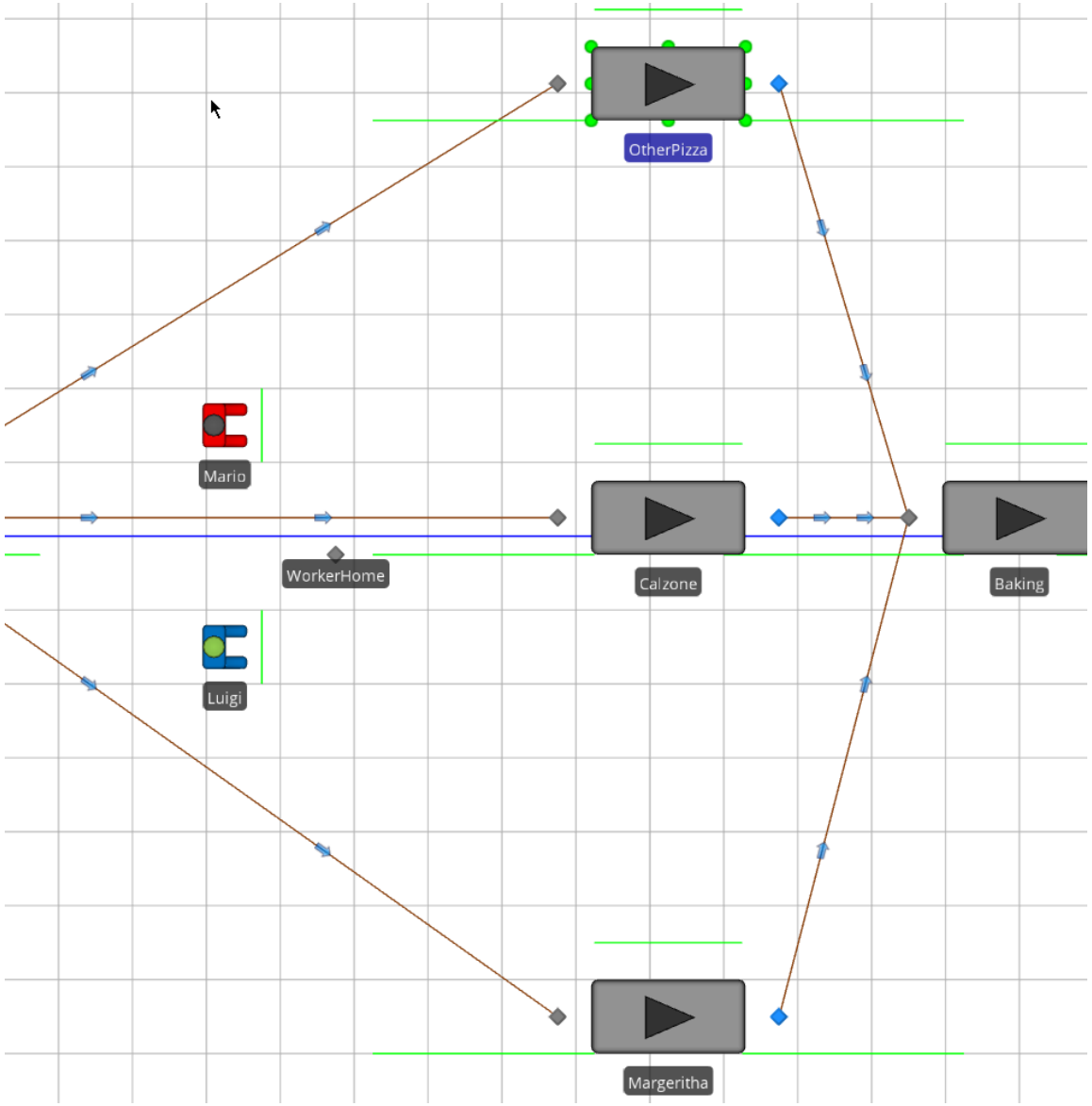
Model Building

Phases of simulation project



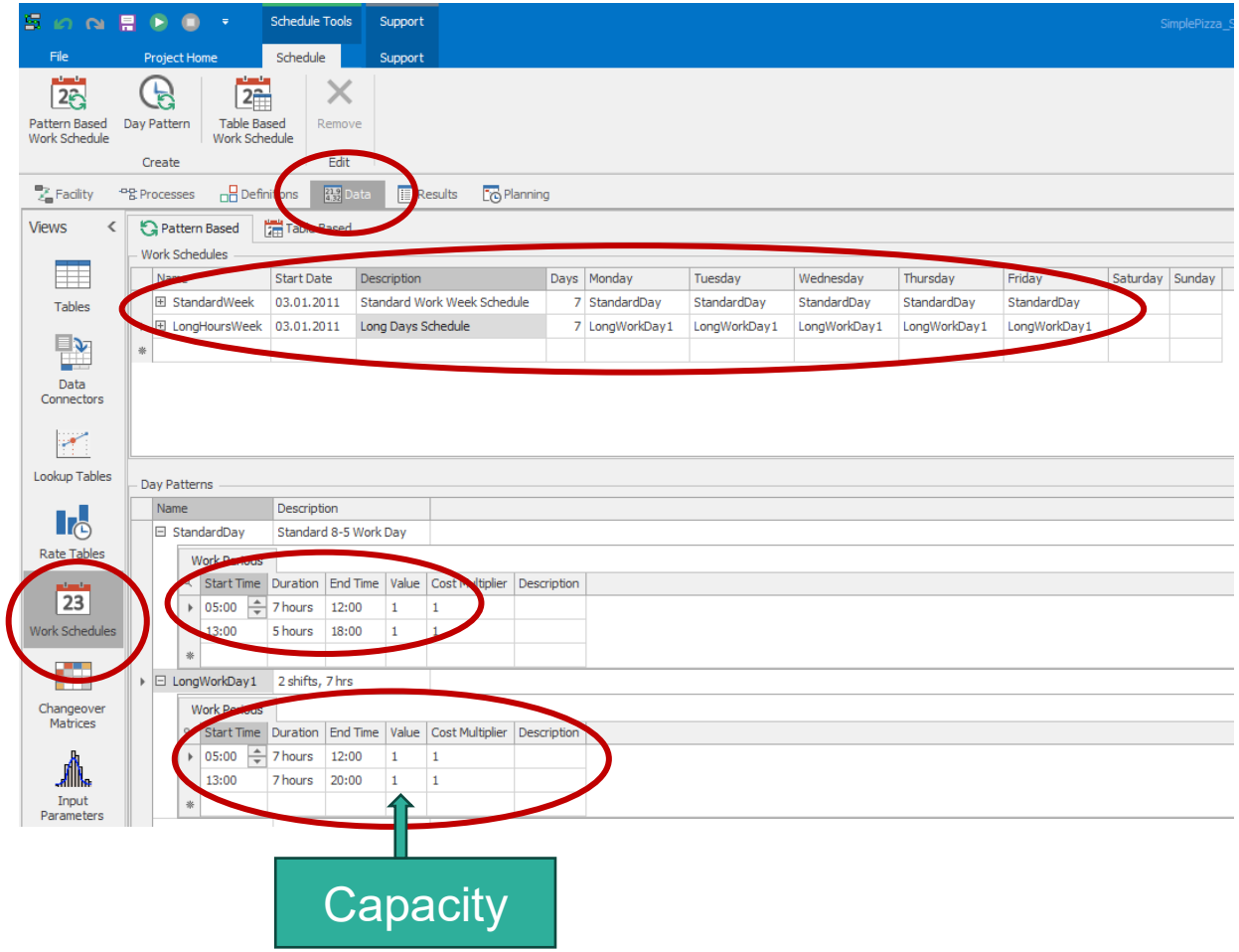
Continue model building

- **New Server for Calzone:**
  - Capacity = 1
  - Process time: Random.Triangular(4,5,6)
- **2 movable workers that work 3 pizza stations**
- **Define own state statistics**
  - Number In System
  - Compare to Simio Stats
- Adding tally statistics
  - Time Between Stations
- Adding TimeSeries Plots and counters

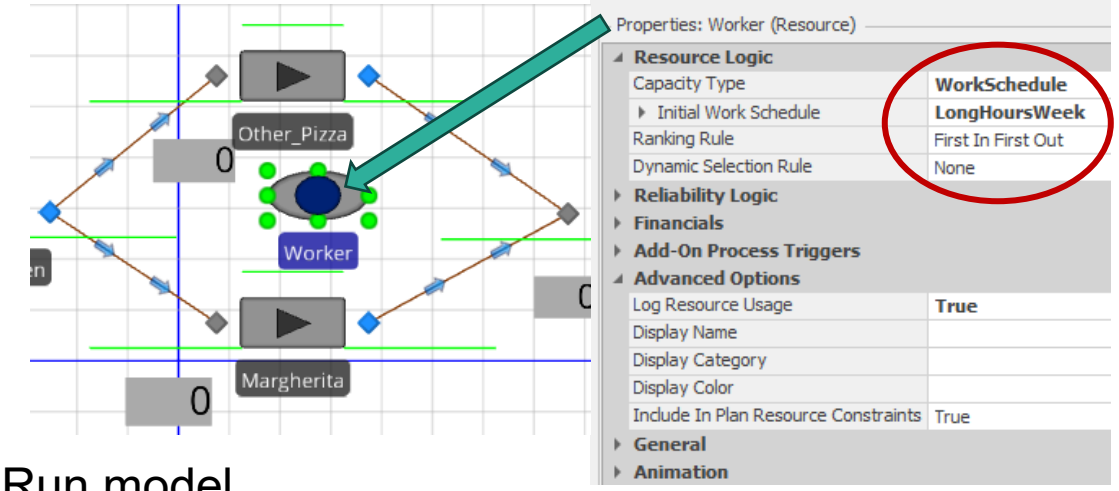


Work Schedules

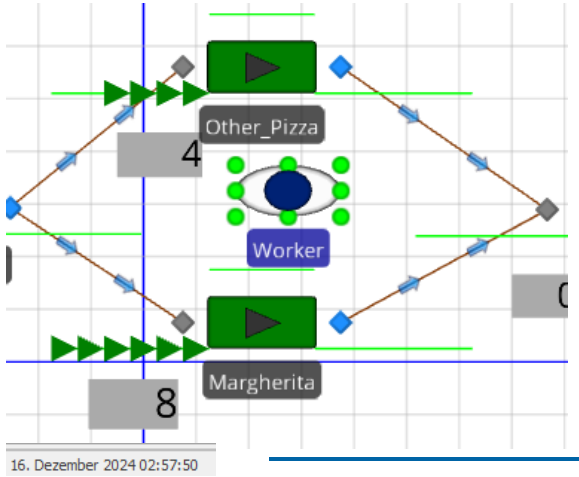
1. Define schedule (one or more)



2. Connect schedule with Server or Resources



3. Run model



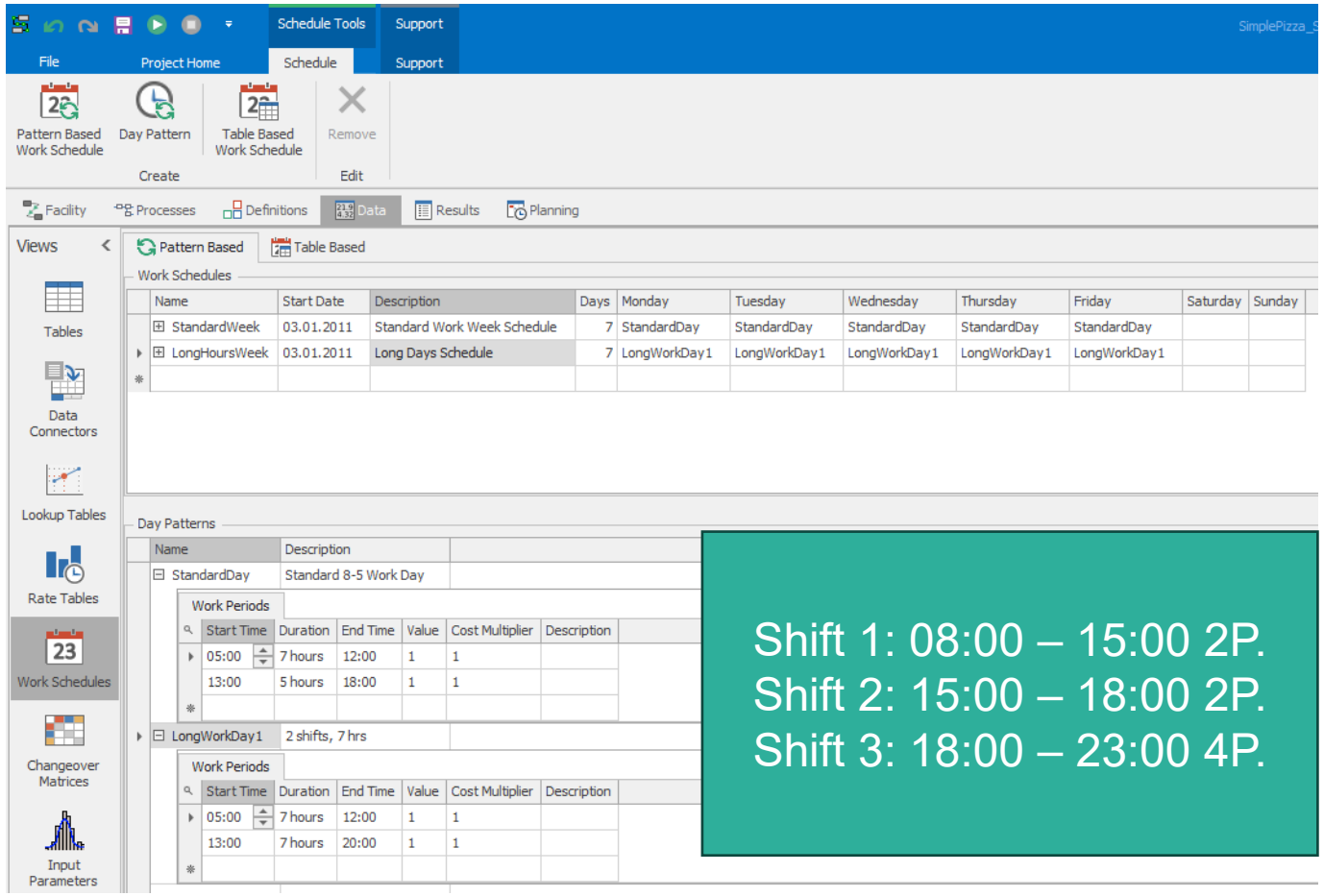
Worker is only available during shift times

Work Schedules

1. Analyse input data

Starting Offset	Ending Offset	Rate (events per hour)
Day 1, 00:00:00	Day 1, 01:00:00	5
Day 1, 01:00:00	Day 1, 02:00:00	5
Day 1, 02:00:00	Day 1, 03:00:00	0
Day 1, 03:00:00	Day 1, 04:00:00	0
Day 1, 04:00:00	Day 1, 05:00:00	0
Day 1, 05:00:00	Day 1, 06:00:00	0
Day 1, 06:00:00	Day 1, 07:00:00	0
Day 1, 07:00:00	Day 1, 08:00:00	0
Day 1, 08:00:00	Day 1, 09:00:00	0
Day 1, 09:00:00	Day 1, 10:00:00	10
Day 1, 10:00:00	Day 1, 11:00:00	15
Day 1, 11:00:00	Day 1, 12:00:00	10
Day 1, 12:00:00	Day 1, 13:00:00	5
Day 1, 13:00:00	Day 1, 14:00:00	5
Day 1, 14:00:00	Day 1, 15:00:00	5
Day 1, 15:00:00	Day 1, 16:00:00	15
Day 1, 16:00:00	Day 1, 17:00:00	50
Day 1, 17:00:00	Day 1, 18:00:00	55
Day 1, 18:00:00	Day 1, 19:00:00	90
Day 1, 19:00:00	Day 1, 20:00:00	100
Day 1, 20:00:00	Day 1, 21:00:00	100
Day 1, 21:00:00	Day 1, 22:00:00	15
Day 1, 22:00:00	Day 1, 23:00:00	10
Day 1, 23:00:00	Day 2, 00:00:00	5

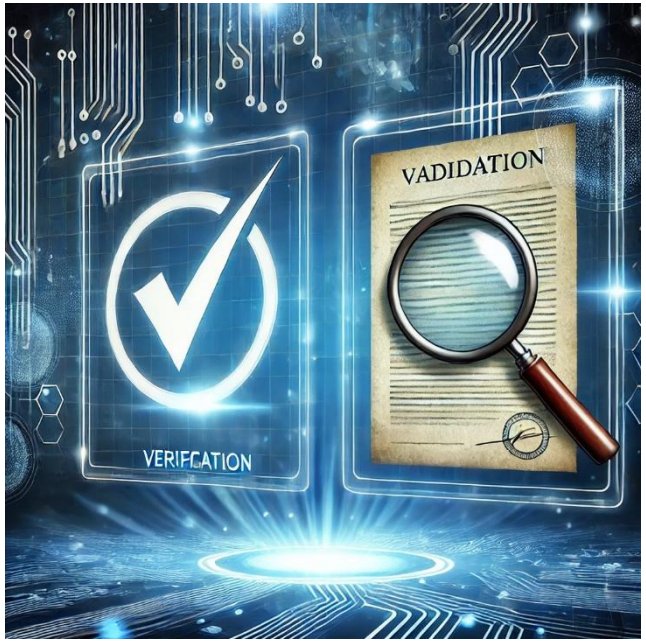
2. Define schedule



Verification and Validation of Simulation Models

Clarification of terms

- **Validation (Validity of the model)**
  - A model is valid if it reflects the structural properties of the real-world system.
  - It must also behave over time like the real system.
  - The process of checking a model's validity is called Validation.
- **Verification (Correctness of the model)**
  - A model is correct when modeling tools (e.g., software or graphical symbols) are used according to the required syntactic rules (syntactic correctness).
  - The model must be free of logical errors (semantic correctness).
  - Verification can be performed independently of the real system.

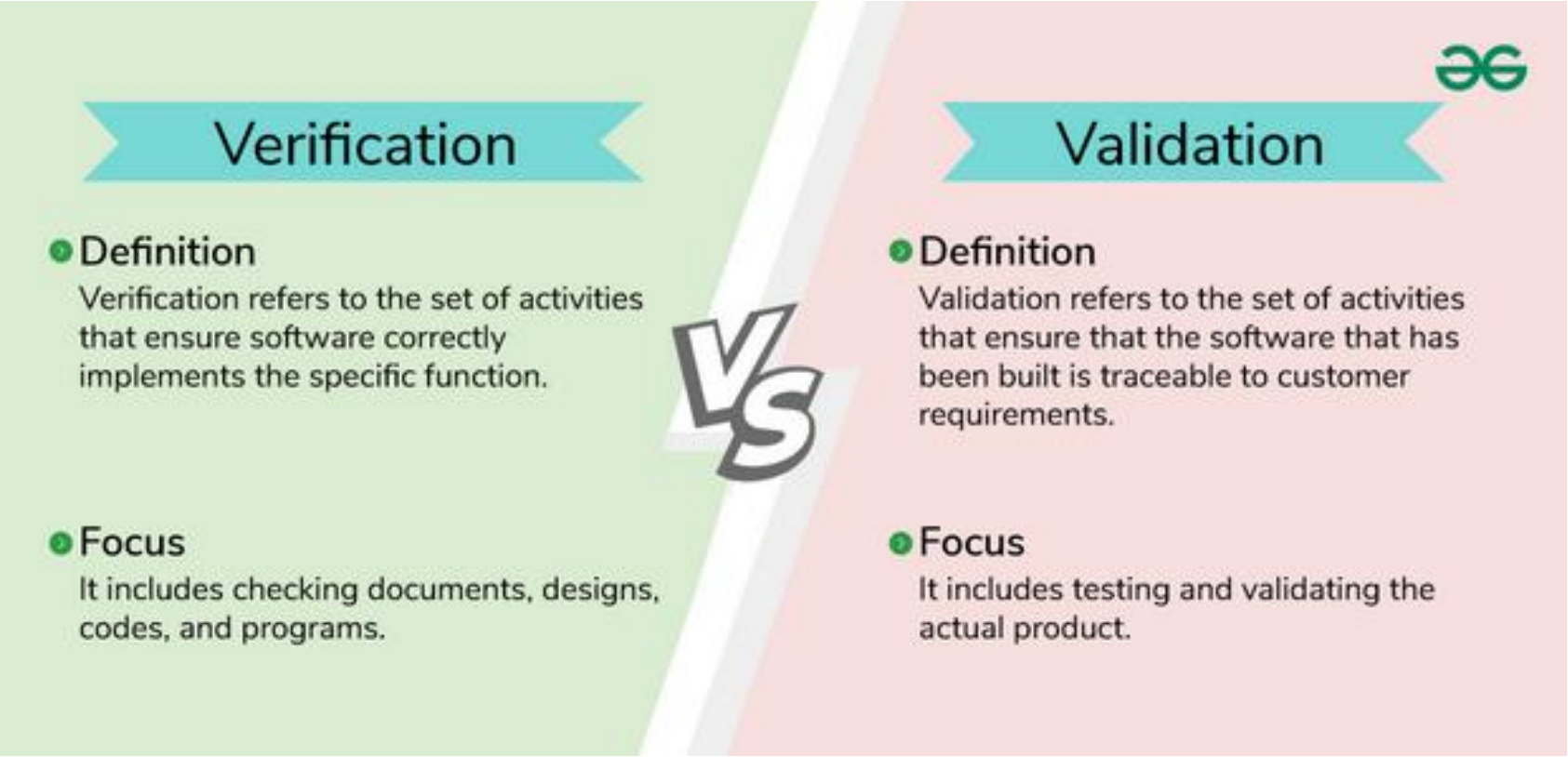


Clarification of terms

Correctness expresses that a model has been created correctly, **validity** means that the correct model has been created.

Source: Hedstück (2013)

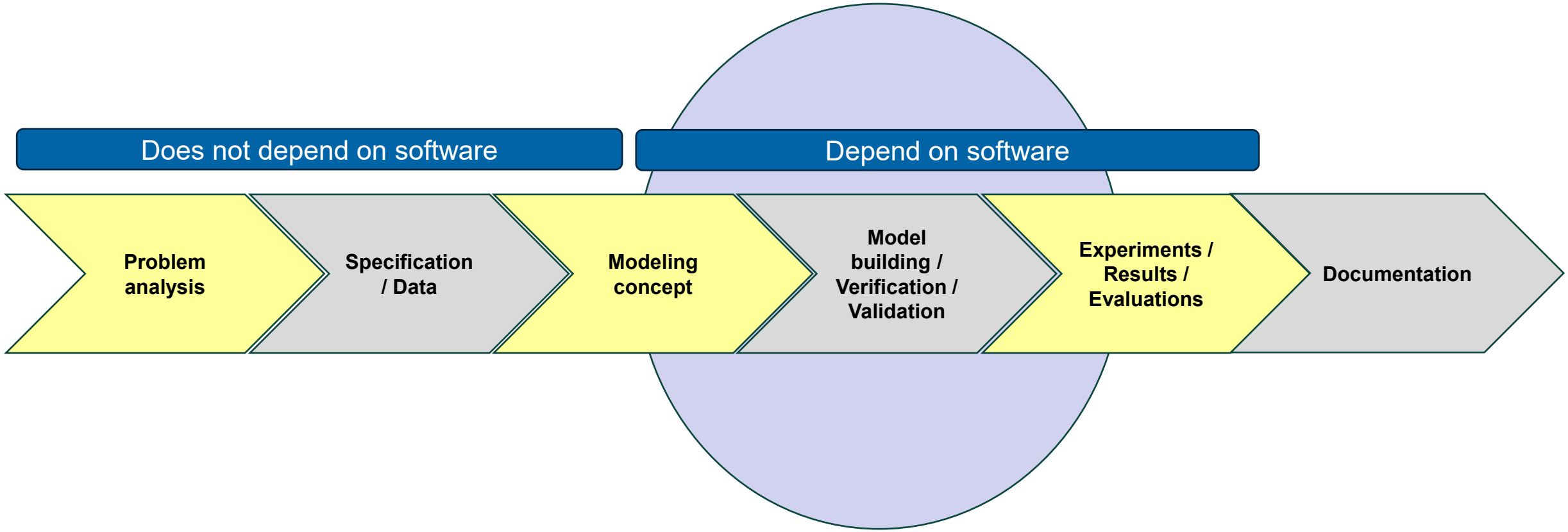
Comparison to software development



Source: https://www.geeksforgeeks.org/differences-between-verification-and-validation/

Validation of a model

Phases of simulation project



- Verification and Validation

Validation of a simulation model

- **Face Validity:** Experts assess whether the model logic and behavior are reasonable for the system being modeled.
- **Degenerate Tests:** The model is tested under extreme or illogical parameter settings to check for appropriate behavior (e.g., queue growth with high arrival rate).
- **Traces:** Tracks individual entities through the model to verify logic and accuracy.
- **Confidence Intervals & Hypothesis Testing:** Statistical tools used to compare model and system outputs, ensuring differences are within acceptable accuracy ranges.
- **Parameter Variability – Sensitivity Analysis:** Analyzes how changes in input parameters affect outputs; sensitive parameters must be estimated accurately.
- **Operational Graphics & Animation:** Graphical outputs or animations of the model help detect unexpected behaviors or inconsistencies.

Validation of a simulation model

- **Validation Is Purpose-Driven:** A model is valid only with respect to its intended purpose. It may be valid for one set of conditions or questions but invalid for others.
- **Validation Is a Process, Not a One-Time Test:** It’s iterative and evolves alongside model development. Multiple rounds of validation are often required as the model is refined.
- **Involvement of Stakeholders**
  Including domain experts and end-users in the validation process increases confidence and relevance.
- **Documentation of Validation:** Keeping detailed and summary records of all validation steps, data sources, techniques used, and conclusions enhances transparency and trust.
- **Validation Failure Is Insightful:** If a model fails validation, it's a valuable diagnostic tool—it points to flaws in data, logic, assumptions, or implementation.

Verification of a model

Verification of a simulation model

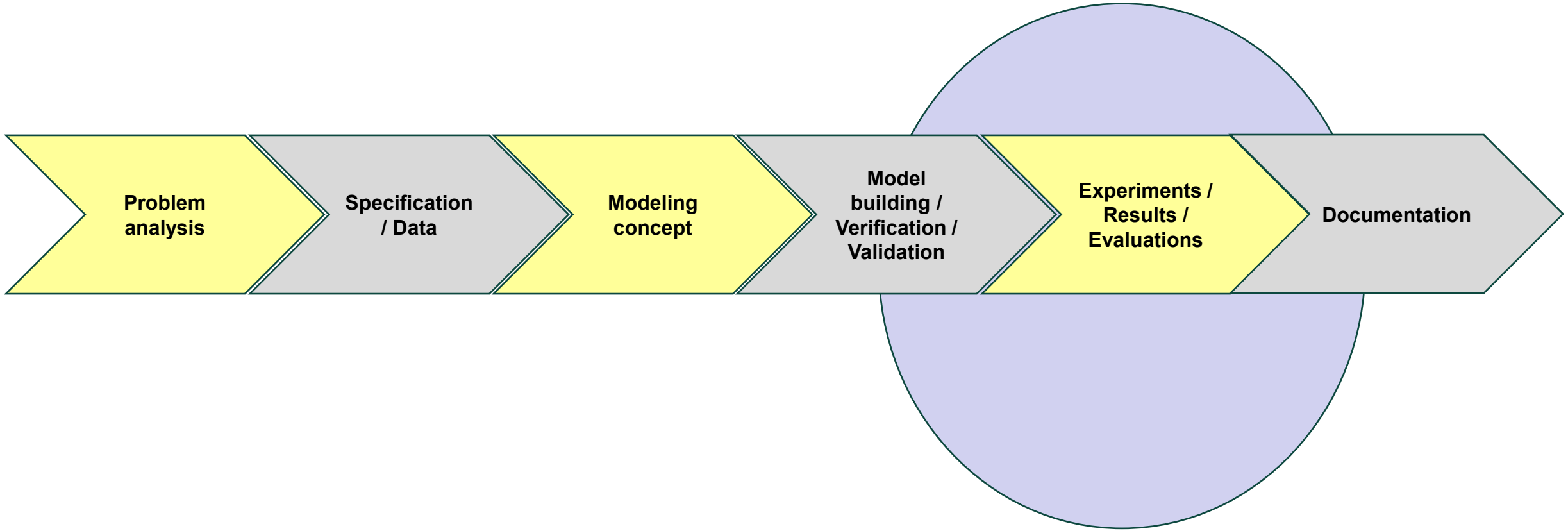
- **Common Techniques**:
  - **Traces**: Follow entities step-by-step through the model to ensure logic is correctly executed.
  - **Structured Walkthroughs**: Systematic manual reviews of code, model logic, or flowcharts to detect errors.
  - **Static Testing**: Analyze the code without running it (e.g., logic inspection, correctness proofs).
  - **Dynamic Testing**: Run the model under controlled conditions and evaluate output and internal states.
  - **Internal Consistency Checks**: Use assertions or rules to detect inconsistencies or logic violations during simulation.
  - **Reprogramming Critical Components**: Implement parts of the model in different ways and compare results.

Verification of a simulation model

- **Common Issues Found in Verification**:
  - Logic errors in process flow
  - Incorrect implementation of event sequencing or timing
  - Bugs in random number generation or distribution sampling
  - Unit mismatches, wrong variable types, indexing errors, etc.
- **Importance of Documentation**
  - All verification activities should be **well-documented** to build confidence and support credibility.

Design of experiments

Phases of simulation project



Possible goals and target values (KPIs)

Objectives:

1. **Proof of performance of the existing system for the future scenarios (increased quantity)**
2. **Is it possible to produce the desired quantity?**
3. **Identification of any bottlenecks and measures to eliminate them**
4. **How many staff are required for production?**
5. How should the system (production capacities and/or processing times) be changed in order to produce the desired quantity?
6. Under what conditions should the orders be produced during the day in order to achieve the desired quantity on the existing or new system?

Target values (KPIs):

1. Throughput times (time in system), waiting times, **number of finished / unfinished orders**
2. Length of Queue, other key figures
3. Utilization of the individual process steps

Experiment planing

The following things should be considered in the design of the experiments:

- What is the target objective of the simulation (e.g. all orders should be finished)
- Actual situation and associated parameters
- How to vary parameters?
- KPIs: How can we "measure" the results?

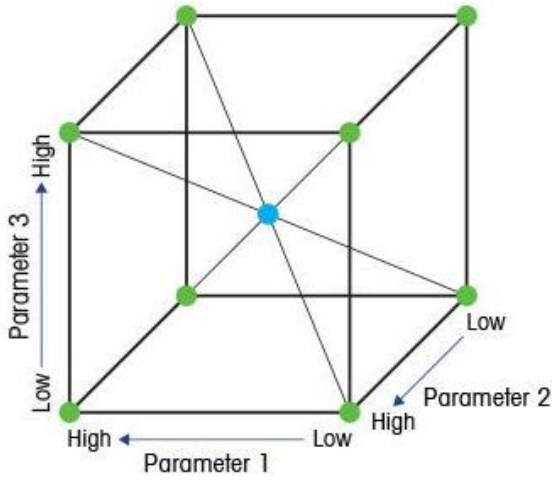
Design of Experiments und Experimentmatrix

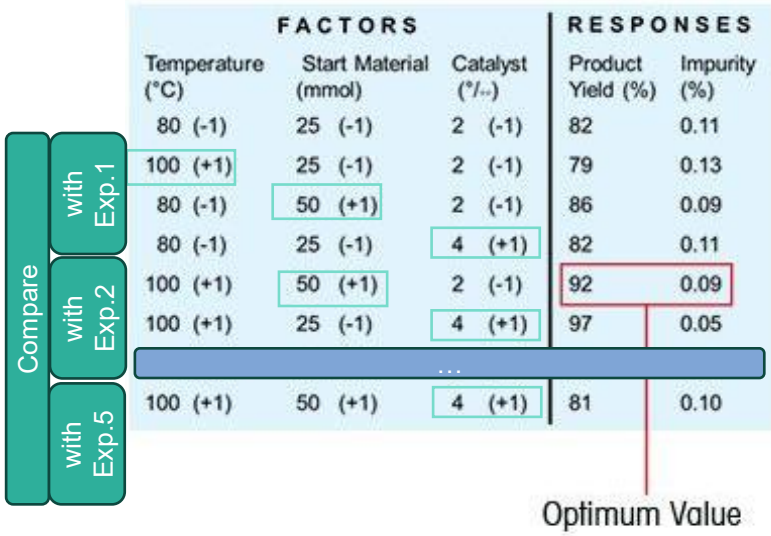
Design of Experiments (DoE) is a statistical approach to reaction and process optimization that allows different factors to be varied simultaneously with the aim of bringing the reaction space to optimal values.

In a Design of Experiments (DoE) study, the factors of interest are systematically varied from their lowest to their highest value and all possible factor combinations are carried out in the same series of experiments.

Advantages:

- Better process understanding in less time
- Shorter development cycles in production





Quelle: https://www.mt.com/ch/en/home/applications/L1_AutoChem_Applications/L2_ReactionAnalysis/design-of-experiments-doe.html

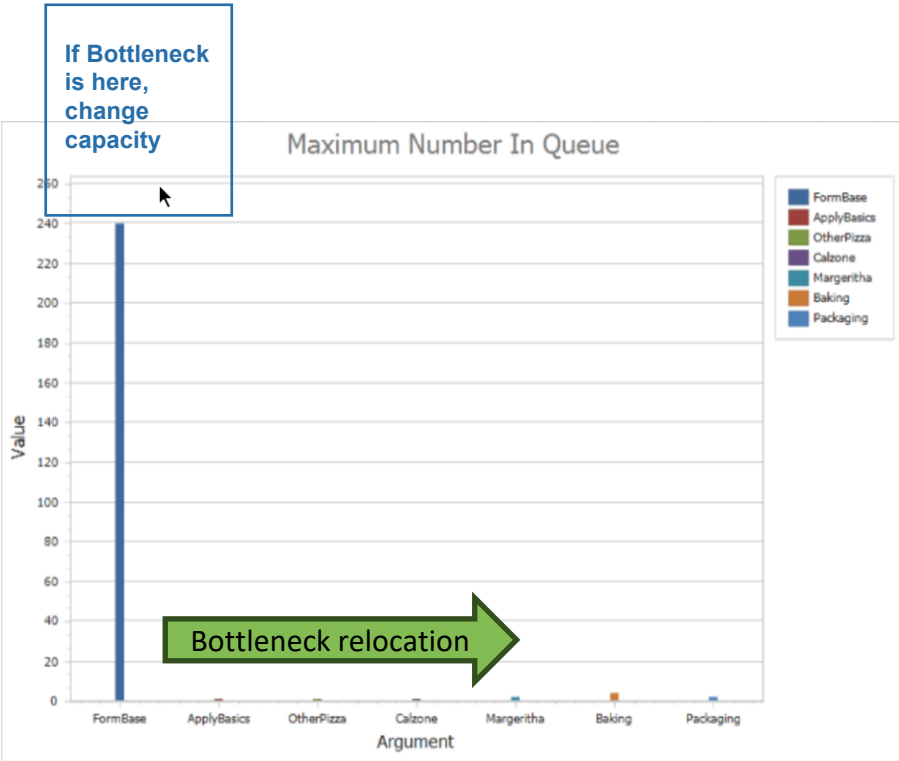
Create experiment matrix

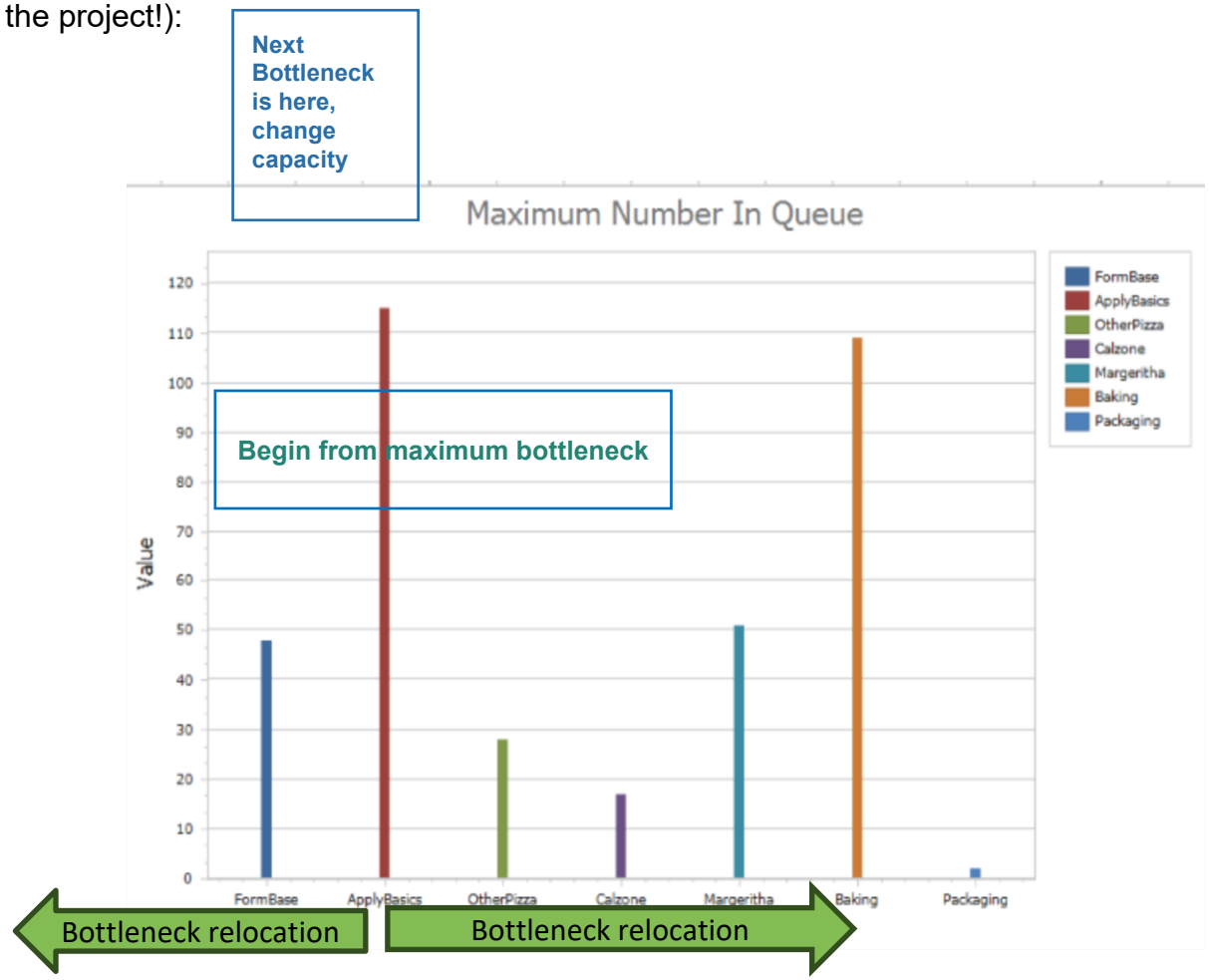
Define the most important parameters (see also the objectives of the project!):

- Capacities, process times, controls, others

Consider how they should be changed:

- At which locations should the parameters be varied?
- Where are the limits of the parameters?





Beispiel 2

Example Data Structure for the Simulation Project

Parameter

Scenario			Replications		Controls					
☑	Name	Status	Required	Completed	OrdersPerDay	FormBase_Capacity	ApplyBasics_Capacity	Baking_Capacity	Packaging_Capacity	Marios_Capacity
☑	Scenario1	Idle	10	0 of 10	500	1	1	3	1	2
☑	Scenario2	Idle	10	0 of 10	500	2	1	3	1	2
☑	Scenario3	Idle	10	0 of 10	500	1	2	3	1	2
☑	Scenario4	Idle	10	0 of 10	500	1	1	3	2	2
☑	Scenario5	Idle	10	0 of 10	500	1	1	3	1	3
☑	Scenario6	Idle	10	0 of 10	500	2	2	3	2	2
☑	Scenario7	Idle	10	0 of 10	500	2	2	3	2	3
☑	Scenario8	Idle	10	0 of 10	500	2	2	3	2	4

ACTUAL Situation

Target Experiments

Create experiment matrix

- Define Experiments:
  - Only change 1 parameter per experiment (otherwise you cannot see which parameter has an influence).
  - If no further effect is apparent, then continue to consider whether there are further measures or whether the investigation is complete.

Design | Response Results | Pivot Grid | Reports | Dashboard Reports | Input Analysis

Scenario			Replications		Controls						
✓	Name	Status	Required	Completed	OrdersPerDay	RateTable	FormBase_Capacity	ApplyBasics_Capacity	Baking_Capacity	Packaging_Capacity	Marios_Capacity
✓	Scenario1	Idle	10	10 of 10	500	OrderRate	1	1	3	1	2
✓	Scenario2	Idle	10	10 of 10	500	OrderRate	2	1	3	1	2
✓	Scenario3	Idle	10	10 of 10	500	OrderRate	1	2	3	1	2
✓	Scenario4	Idle	10	10 of 10	500	OrderRate	1	1	3	2	2
✓	Scenario5	Idle	10	10 of 10	500	OrderRate	1	1	3	1	3
✓	Scenario6	Idle	10	10 of 10	500	OrderRate	2	2	3	2	2
✓	Scenario7	Idle	10	10 of 10	500	OrderRate	2	2	3	2	3
✓	Scenario8	Idle	10	10 of 10	500	OrderRate	2	2	3	2	4

Documentation

Projektphasen

Modeling: Assignment 1

Experiment: Assignment 2

Presentation next week

Documentation (short report)

Documentation of Project Assignment 2

Contents:

- **Cover page and table of contents**
- **Introduction (0.5 pages):**
  - Presentation of the assignment (experimental design and evaluation)
  - Objective: What is to be achieved by the simulation and the experiments?
  - Brief overview of the simulated process and the simulation environment used
- **Experiment planning matrix (1-2 pages):**
  - Presentation of the matrix (e.g. as a table)
  - Explanation of the selected scenarios and parameter changes
  - Justification of the series of experiments: Why were these specific scenarios chosen?
- **Evaluation (1-2 pages):**
  - Presentation of the results using diagrams and tables
  - Analysis of the predefined KPIs and stochastic elements
  - Interpretation of the results: What findings can be derived?
- **Discussion and conclusions (1-2 pages):**
  - Summary of the most important results
  - Derivation of well-founded conclusions for the simulated process
  - Outlook: Possible further questions or future test series

Evaluation Grid (on Moodle)

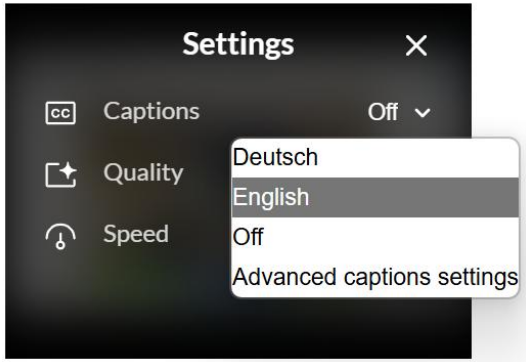
Presentation (Assignment 1)		
Evaluation criteria		Max. Punkte
Creativity and originality		6
The idea is innovative and creatively implemented.		3
The visualizations are appealing and understandable.		3
Comprehensibility of the process description		6
The process is explained in a clear and comprehensible way.		3
The representations (BPMN diagram or other graphics) support the understanding of the process.		3
Presentation of parameters and KPIs		6
The most important parameters and KPIs are clearly named and explained.		3
The importance of KPIs for process evaluation is understandable.		3
Stochastic elements and simulation		6
Uncertainties and random factors in the process are explained in an understandable way.		3
It will clearly describe how these stochastic elements could be incorporated into the simulation?		3
Quality of presentation		12
The presentation is sensibly structured and has a logical flow.		4
Clear and understandable language is used. It is spoken clearly and freely (without reading).		4
The time is well used and adhered to. The pace of the proceedings is pleasant.		4
Total		36

Documentation (Assignment 2)		
Evaluation criteria		Max. Punkte
Structure and design		4
The work is formatted correctly		2
The structure is clear and comprehensible.		2
Contents		16
Content is consistent with the task and the objective.		2
The procedure is described.		2
The initial situation and goals, input data, experiment matrix or key figures / evaluations are sensibly selected (suitable for the task) and justified.		5
Explanations of the diagrams and tables are available, correctly described and comprehensible.		5
Summary / discussion available and useful		2
Presentation (images, tables and diagrams)		8
Diagrams or illustrations match the content.		3
Figures and diagrams correctly described and easy to read (titles, axes, etc.).		2
Tables are structured in a meaningful way. The contents of the tables are correct.		3
Language		8
The wording is clear and comprehensible. The statements are clearly formulated. The language is appropriate to the topic, fluent and easy to understand. The technical terms are explained.		4.0
The texts are grammatically correct and error-free.		4.0
Total		36

At home

How does a simulation study work?

- Additional information and learning control in Moodle course
- **Module 3: at home till next Thursday**

Videos are in German, but have English subtitles

To switch on: